LAB SESSION 05

ALGORITHM TO DELETE THE FIRST NODE FROM THE DOUBLY LINKED LIST

Step 1: IF START = NULL, then
Write UNDERFLOW
Go to Step 6
[END OF IF]

Step 2: SET PTR = START
Step 3: SET START = START → NEXT
Step 4: IF START ≠ NULL, then
SET START → PREV = NULL
[END OF IF]
Step 5: FREE PTR
Step 6: EXIT

ALGORITHM TO DELETE THE LAST NODE OF THE DOUBLY LINKED LIST

Step 1: IF START = NULL, then
Write UNDERFLOW
Go to Step 9
[END OF IF]

Step 2: SET PTR = START
Step 3: Repeat Step 4 while PTR → NEXT ≠ NULL
Step 4: SET PTR = PTR → NEXT
[END OF LOOP]

Step 5: SET PREPTR = PTR → PREV
Step 6: SET PREPTR → NEXT = NULL
Step 7: FREE PTR
Step 8: EXIT

ALGORITHM TO DELETE THE NODE AFTER A GIVEN NODE FROM THE DOUBLY LINKED LIST

Step 1: IF START = NULL, then
    Write UNDERFLOW
    Go to Step 12
[END OF IF]

Step 2: SET PTR = START
Step 3: Repeat Step 4 while PTR → DATA ≠ NUM
Step 4: SET PTR = PTR → NEXT
[END OF LOOP]

Step 5: SET TEMP = PTR → NEXT
Step 6: IF TEMP = NULL, then
    Go to Step 12
[END OF IF]

Step 7: SET PTR → NEXT = TEMP → NEXT
Step 8: IF TEMP → NEXT ≠ NULL, then
    SET TEMP → NEXT → PREV = PTR
[END OF IF]

Step 9: FREE TEMP
Step 10: EXIT

TASKS:

1. Modify deleteAtPosition() to handle deletion from the last position without calling deleteAtEnd().
2. Implement a function to reverse a doubly linked list.
3. Implement a function to merge two DLLs.
4. Create a menu-driven program to test all operations dynamically.